Week 4

Topic: Divide and Conquer

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240. Search a 2D Matrix II (Medium)

Write an efficient algorithm that searches for a value target in an m x n integer matrix matrix. This matrix has the following properties:

- Integers in each row are sorted in ascending from left to right.
- Integers in each column are sorted in ascending from top to bottom.

Example 1:

Input: matrix = [[1,4,7,11,15],[2,5,8,12,19],[3,6,9,16,22],[10,13,14,17,24],[18,21,23,26,30]], target = 5

Output: true

For detailed descriptions and constraints, please refer to Leetcode 240.

218. The Skyline Problem (Hard)

A city's **skyline** is the outer contour of the silhouette formed by all the buildings in that city when viewed from a distance. Given the locations and heights of all the buildings, return *the **skyline** formed by these buildings collectively*.

The geometric information of each building is given in the array buildings where buildings[i] = [$left_i$, $right_i$, $height_i$]:

- $left_i$ is the x coordinate of the left edge of the ith building.
- $right_i$ is the x coordinate of the right edge of the ith building.
- $height_i$ is the height of the ith building.

You may assume all buildings are perfect rectangles grounded on an absolutely flat surface at height 0.

The **skyline** should be represented as a list of "key points" **sorted by their x-coordinate** in the form [[x_1,y_1],[x_2,y_2],...]. Each key point is the left endpoint of some horizontal segment in the skyline except the last point in the list, which always has a y-coordinate 0 and is used to mark the skyline's termination where the rightmost building ends. Any ground between the leftmost and rightmost buildings should be part of the skyline's contour.

Note: There must be no consecutive horizontal lines of equal height in the output skyline. For instance, [...,[2 3],[4 5],[7 5],[11 5],[12 7],...] is not acceptable; the three lines of height 5 should be merged into one in the final output as such: [...,[2 3],[4 5],[12 7],...]

Example 1:

Input: buildings = [[2,9,10],[3,7,15],[5,12,12],[15,20,10],[19,24,8]]

Output: [[2,10],[3,15],[7,12],[12,0],[15,10],[20,8],[24,0]]

Explanation:

Figure A shows the buildings of the input.

Figure B shows the skyline formed by those buildings. The red points in figure B represent the key points in the output list.

For detailed descriptions and constraints, please refer to Leetcode 218.

761. Special Binary String (Hard)

Special binary strings are binary strings with the following two properties:

- The number of 0's is equal to the number of 1's.
- Every prefix of the binary string has at least as many 1's as 0's.

You are given a **special binary** string s.

A move consists of choosing two consecutive, non-empty, special substrings of s, and swapping them. Two strings are consecutive if the last character of the first string is exactly one index before the first character of the second string.

Return *the lexicographically largest resulting string possible after applying the mentioned operations on the string*.

Example 1:

Input: s = "11011000"

Output: "11100100"

Explanation: The strings "10" [occuring at s[1]] and "1100" [at s[3]] are swapped.

This is the lexicographically largest string possible after some number of swaps.

For detailed descriptions and constraints, please refer to Leetcode 761.